
Counting and conditional probability

Events · Counting · Bayes' theorem

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Probability and Statistics for Engineers

Three lectures of 65 minutes

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Two coin tosses

Experiment: toss a fair coin twice and record the results in order.

Model: the first result does not change the chances on the second toss.

This is the **independence** assumption; each outcome below has probability $1/4$.

HH

HT

TH

TT

$$\Omega = \{HH, HT, TH, TT\}. \quad HT \neq TH$$

What is the probability of exactly one head?

First write the event as a set of outcomes. Then add their probabilities.

Events in words and symbols

First toss heads: $A = \{HH, HT\}$. Second toss heads: $B = \{HH, TH\}$.

In words	Event	Outcomes
Both tosses are heads	$A \cap B$	$\{HH\}$
At least one head	$A \cup B$	$\{HH, HT, TH\}$
First toss is not heads	A^c	$\{TH, TT\}$
Second heads, first not	$B \setminus A = B \cap A^c$	$\{TH\}$
Neither toss is heads	$(A \cup B)^c = A^c \cap B^c$	$\{TT\}$

Check: is “exactly one head” the same as $A \cup B$?

Events are disjoint when they share no outcomes: $A \cap B = \emptyset$.

Basic probability rules

A probability model satisfies

- $P(A) \geq 0$ for every event.
- $P(\Omega) = 1$.
- Add probabilities for disjoint events:

$$P(E \cup F) = P(E) + P(F).$$

Additivity also holds for a finite or countably infinite collection of pairwise disjoint events.

When outcomes are equally likely

$$P(A) = \frac{|A|}{|\Omega|}.$$

Here $|A|$ means the number of outcomes in A ; Ω is finite.

For two fair, independent tosses,

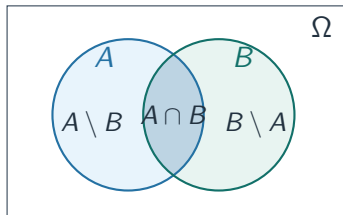
$$P(\text{at least one head}) = \frac{3}{4}.$$

Complement: all outcomes outside the event

A and A^c are disjoint and cover Ω , so $P(A^c) = 1 - P(A)$.

The union rule

How do we find the probability that A or B occurs?



Adding $P(A)$ and $P(B)$ counts the overlap twice. Subtract it once.

More formally, split B into disjoint pieces:

$$P(B) = P(A \cap B) + P(B \setminus A).$$

Then add A and the part of B outside A .

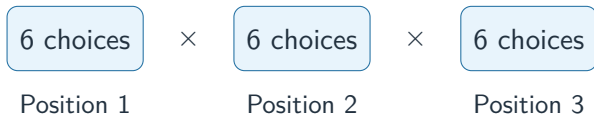
For any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Two tosses: use first-heads A and second-heads B .

Count one choice at a time

A robot sends a three-symbol code. Each position can use any of six symbols.



How many codes are possible if repetition is allowed?

Counting successive choices

Multiply the choices at each stage: $n_1 n_2 \cdots n_k$, provided stage i has n_i choices after *every* allowed earlier sequence.

k ordered choices from n objects, with repetition allowed: n^k .

Counting outcomes alone says nothing about their probabilities.

Ordered selections without replacement

From ten people, assign a president, a secretary, and a treasurer.
Each person can hold only one role.

President	Secretary	Treasurer
10 choices	9 remaining	8 remaining

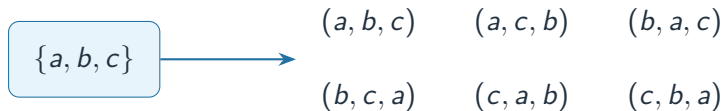
Choose k distinct objects from n , in order

$$(n)_k = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

$n! = n(n-1) \cdots 1$; $0! = 1$. For integers $0 \leq k \leq n$, $(n)_0 = 1$.

One committee, six orders

A committee has no roles. Reordering its members does not change the committee.



Why divide by 3!?

Each three-person committee appears in $3 \cdot 2 \cdot 1 = 6$ ordered lists.

More generally, each k -element subset appears in exactly $k!$ orders.
Division works because every subset is counted the same number of times.

Unordered selections

Return to the ten people: how many committees of three can we choose?

Combinations: choose k objects from n , without order

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!(n-k)!}, \quad 0 \leq k \leq n.$$

The counting argument

$$\underbrace{(n)_k}_{\text{ordered lists}} = \underbrace{\binom{n}{k}}_{\text{subsets}} \times \underbrace{k!}_{\text{orders per subset}}.$$

“ n choose k ” counts subsets. It is a count, not a probability.

Four jobs, six servers

Assign four labeled jobs to six servers.

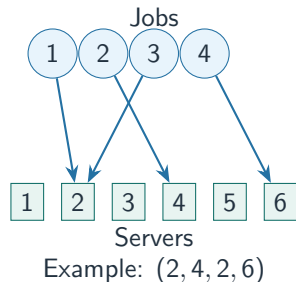
Model assumptions

- Each job chooses any server with probability $1/6$.
- One job's choice does not affect the others.

Record servers in job order:

$$\omega = (x_1, x_2, x_3, x_4).$$

Each $x_i \in \{1, \dots, 6\}$; each tuple has probability $1/6^4$.



What is the probability of a collision?

A collision means at least two jobs choose the same server.

Illustrative allocation model; no performance claim about a real system.

A collision or no collision

Question: what is the probability that at least two jobs share a server?

$C = \{\omega : \text{at least two of } x_1, x_2, x_3, x_4 \text{ are equal}\},$

$C^c = \{\omega : \text{all four server choices are different}\}.$

All outcomes

Six choices per job.

No collision

6, 5, 4, then 3 choices.

Use equal probabilities, then take the complement.

Practice: at least one marked item

Choose a three-item subset from ten distinct items, with all such subsets equally likely. Four items are marked; six are unmarked.



What is the probability of choosing at least one marked item?

First identify the outcomes. Then consider the event “none marked.”

Counting summary

Outcome	Repetition	Count
Ordered length- k sequence	Allowed	n^k
Ordered length- k sequence	Not allowed	$(n)_k = n!/(n-k)!$
Unordered k -element subset	Not allowed	$\binom{n}{k} = n!/[k!(n-k)!]$

$n \geq 1$ and $k \geq 0$ are integers. Without replacement, $k \leq n$.

Before dividing counts

Specify the outcomes and justify equal probabilities: $P(A) = |A|/|\Omega|$.

Two useful event rules

$$P(A^c) = 1 - P(A), \quad P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Wednesday: conditional probability

Today: calculate probabilities after learning part of an outcome.

Warm-up: two dice

Roll two independent fair dice. Record the first and second die in order.
The $6 \cdot 6 = 36$ ordered pairs are equally likely.

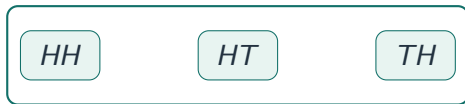
What is the probability that the sum is 7?

Let A be the event “the sum is 7.” List its outcomes first.

Later: what changes if we learn that at least one die is 4?

Conditional probability

Given at least one head in two fair, independent tosses, how likely is exactly one head?



Keep B : at least one head



Exclude

Condition: $B = \{HH, HT, TH\}$. Event of interest: $A = \{HT, TH\}$.

Keep the outcomes in B ; rescale their probabilities to total 1

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Conditioning with unequal probabilities

A request is sent to one of three servers with the probabilities below.

Given that it did not go to server 1, how likely is server 2?

Server	Original probability	Given server 2 or 3
1	1/2	0 (excluded)
2	1/3	_____
3	1/6	_____

$\Omega = \{1, 2, 3\}$; keep $B = \{2, 3\}$. The question asks for $A = \{2\}$.

Count retained outcomes only when their probabilities are equal.

Two dice: at least one is four

		Second die					
		1	2	3	4	5	6
First die	1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
	2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
	3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
	4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
	5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
	6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

Given at least one die is 4, how likely is a sum of 7?

$A = \{\text{sum is 7}\}.$

$B = \{\text{at least one die is 4}\}.$

Outlined row and column: keep B .

Circled cells: $A \cap B$.

Different conditions, different probabilities

Keep the same two fair, independent dice and $A = \{\text{sum is } 7\}$.

Condition B	$ B $	$ A \cap B $	$P(A B)$
At least one die is 4	11	2	2/11
First die is 4	_____	_____	_____
Exactly one die is 4	_____	_____	_____

Explain the differences

- Which condition excludes $(4, 4)$?
- Which condition excludes $(3, 4)$?

The conditional product rule

Same dice example: a sum of 7 *and* at least one die equal to 4.

$$\underbrace{\frac{11}{36}}_{P(B)} \times \underbrace{\frac{2}{11}}_{P(A|B)} = \underbrace{\frac{2}{36}}_{P(A \cap B)}.$$

Multiply the chance of B by the chance of A within B

$$P(A \cap B) = P(B) P(A | B), \quad P(B) > 0.$$

Why it works: multiply both sides of the conditional definition by $P(B)$.

If $P(A) > 0$, the same intersection is $P(A) P(B | A)$.

The rule works without independence: keep the condition in the second factor.

Complements within a condition

Two fair, independent tosses. We know there is at least one head:

$$B = \{HH, HT, TH\}.$$

Exactly one head

$$A = \{HT, TH\}.$$

$$P(A \mid B) = \frac{2}{3}.$$

Not exactly one head

Within B , only HH remains.

$$P(A^c \mid B) = 1 - \frac{2}{3} = \frac{1}{3}.$$

Probability rules still apply after conditioning

For $P(B) > 0$,

$$P(B \mid B) = 1, \quad P(A^c \mid B) = 1 - P(A \mid B).$$

$A \cap B$ and $A^c \cap B$ are disjoint and together make up B .

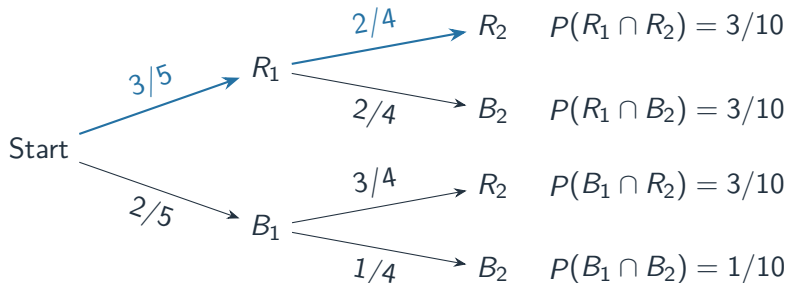
Keep the same condition on both sides of the complement rule.

Two draws without replacement

A bag contains three red and two blue balls. Draw twice without replacement.

At each draw, every remaining ball is equally likely.

Let R_i mean red on draw i and B_i mean blue on draw i .



Each branch is conditional on the earlier draw. Multiply along a complete path.

Two red draws

What is the probability that both draws are red?

	Before drawing	After one red draw
Red balls	3	2
All remaining balls	5	4
Chance of red next	$3/5$	$2/4$

The second probability describes the remaining balls

The first ball is not replaced, so the second factor is $2/4$, not $3/5$.

Three draws and the chain rule

What is the probability of red, then red, then blue?

Use the same bag and draw three times without replacement.

First: red	After red: red	After two reds: blue
3 red of 5 balls	2 red of 4 balls	2 blue of 3 balls

Apply the product rule twice

$$\begin{aligned}P(A \cap B \cap C) &= P(A \cap B) P(C \mid A \cap B) \\&= P(A) P(B \mid A) P(C \mid A \cap B).\end{aligned}$$

Assume $P(A \cap B) > 0$. Each factor uses all the earlier information.

Practice: both draws or a given first draw

Use the same bag: three red and two blue balls; no replacement.

Two different questions

1. What is the probability that the two draws have different colors?
2. Given the first draw is blue, what is the probability the second is red?

“Given blue first” starts after that draw. “Blue then red” includes reaching it.

Conditional probability summary

For $P(B) > 0$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(B) P(A \mid B).$$

1. Name the known event B in words.
2. Find the outcomes in both A and B .
3. Divide by the probability of the condition B .

Direction matters

$P(A \mid B)$ and $P(B \mid A)$ usually answer different questions.

Equal outcome probabilities allow counts. Otherwise, use their probability weights.
Friday: combine disjoint cases, then reverse the conditioning.

Friday: total probability and Bayes

A device comes from one of several suppliers. We want to answer two questions.

Before testing the device

How likely is a defect?

Combine the suppliers' contributions.

Warm-up: write $P(A \cap B)$ in two ways, using conditional probabilities.

After finding a defect

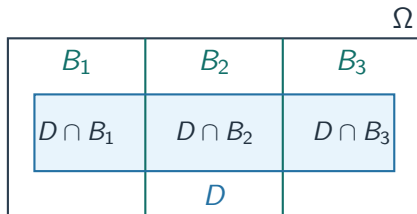
Which supplier is more likely?

Update using the observed defect.

Partitions

Each device comes from exactly one of three suppliers.

Let $B_i = \{\text{supplier } i\}$ and $D = \{\text{device is defective}\}$.



Schematic: region areas do not represent probabilities.

No overlap

A device has only one supplier.

No missing cases

Every device has a supplier.

These cases form a **partition** of Ω .

Definition

$B_1, \dots, B_m$ partition Ω when $B_i \cap B_j = \emptyset$ for $i \neq j$
and $B_1 \cup \dots \cup B_m = \Omega$.

The law of total probability

A defect can come from supplier 1, 2, or 3. How do we combine these cases?

The pieces $D \cap B_1$, $D \cap B_2$, and $D \cap B_3$ are disjoint.

$$\begin{aligned} P(D) &= P(D \cap B_1) + P(D \cap B_2) + P(D \cap B_3) \\ &= P(D \mid B_1) P(B_1) + P(D \mid B_2) P(B_2) \\ &\quad + P(D \mid B_3) P(B_3). \end{aligned}$$

General rule: sum over a partition

$$P(A) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

Assume the B_i partition Ω and $P(B_i) > 0$.

Weight each conditional probability by how likely its case is.

Parts with probability zero contribute zero and can be left out of the conditional sum.

Defects from three suppliers

Select a device at random. Its supplier shares and defect rates are:

Supplier	Share $P(B_i)$	Defect rate $P(D B_i)$	$P(D \cap B_i)$
1	0.20	0.01	_____
2	0.30	0.02	_____
3	0.50	0.03	_____

What is the probability that the selected device is defective?

Use the supplier shares as weights: they do not supply equal numbers of devices.

Bayes' theorem

Given an observed event A , how likely is case B_j ?

Start with the conditional definition and expand its numerator:

$$P(B_j | A) = \frac{P(A \cap B_j)}{P(A)} = \frac{P(A | B_j) P(B_j)}{P(A)}.$$

Use total probability for the denominator

$$P(B_j | A) = \frac{P(A | B_j) P(B_j)}{\sum_{i=1}^m P(A | B_i) P(B_i)}.$$

Numerator

Case B_j and observation A occur together.

B_i form a partition; $P(B_i) > 0$ and $P(A) > 0$.

$P(B_j)$ is the prior; $P(B_j | A)$ is the posterior, after observing A .

Denominator

All ways the observation A can occur.

Which supplier made the defective device?

Given a defect, what is the probability the device came from supplier 3?

The previous contributions, shown on a common scale of 1,000 devices:

	Supplier 1	Supplier 2	Supplier 3
All devices	200	300	500
Defective devices	2	6	15

Counts represent the model proportions, not a guaranteed observed sample.

Restrict to defective devices. There are $2 + 6 + 15 = 23$ on this scale.

Before observing the defect: $P(B_3) = 50\%$.

After observing it: compare supplier 3 with all defective devices.

A rare issue and an alert

A monitoring system checks devices for a fault. Use this illustrative model:

$G = \{\text{genuine fault}\}$; $L = \{\text{alert}\}$.

What the model tells us	Probability
A device has a fault	$P(G) = 0.01$
Alert, given a fault	$P(L \mid G) = 0.90$
Alert, given no fault	$P(L \mid G^c) = 0.05$

Given an alert, how likely is a genuine fault?

We want $P(G \mid L)$. This reverses the condition in $P(L \mid G)$.

Illustrative probabilities; no company or measured dataset is being described.

Count the devices with an alert

Show the model proportions on a scale of 10,000 devices.

	Alert L	No alert L^c	Total
Fault G	90	10	100
No fault G^c	495	9,405	9,900
Total	585	9,415	10,000

Given an alert: keep the alert column. Which of those devices have a fault?

The fault rate is only 1%. The much larger no-fault group produces many false alerts. These counts represent model proportions; actual observed counts may vary.

Practice: which box was chosen?

First choose a box. Then choose one of its four balls uniformly.

	Chance of choosing box	Contents	Chance of red
Box 1	1/3	3 red, 1 blue	3/4
Box 2	2/3	1 red, 3 blue	1/4

Given a red draw, what is the probability we chose box 1?

Let $B_i = \{\text{box } i\}$ and $R = \{\text{red draw}\}$. We want $P(B_1 | R)$.

Total probability and Bayes: summary

Total probability: combine all cases

$$P(A) = \sum_{i=1}^m P(A \mid B_i) P(B_i).$$

Bayes: update a case after an observation

$$P(B_j \mid A) = \frac{P(A \mid B_j) P(B_j)}{P(A)}.$$

Before calculating

- The cases must be disjoint and cover the sample space.
- Use each case's probability as its weight.
- Conditioning events must have positive probability.

Next week: independence—when does conditioning leave a probability unchanged?

References

DeGroot and Schervish

Probability and Statistics, fourth edition.

- Sections 1.4–1.8 and 1.10: sets, probability rules, counting, and unions.
- Section 2.1: conditional probability, product rule, and total probability.
- Section 2.3: Bayes' theorem; Example 2.3.4 motivates the supplier example.

Mor Harchol-Balter

Introduction to Probability for Computing (2024).

- Section 1.5: permutations and combinations.
- Sections 2.1–2.3, 2.5–2.6: events, probability, conditioning, and Bayes.

Additional background: Sheng Jiang, *STAT 131: Introduction to Probability Theory*, July 18, 2024, pp. 10–19; Antonio de Leon's earlier STAT 131 discussion sheets and recap slides. Textbook reading is complementary; slides and course notes are the primary materials.